

The $z \simeq 4\text{--}6$ Massive-Galaxy Abundance is Consistent with IllustrisTNG Once the Stellar-Mass Systematic Budget is Accounted For

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¹*Descriptive, machine-generated draft — not a validated measurement.*

ABSTRACT

The reported over-abundance of massive galaxies at $z > 4$ relative to Λ CDM hydrodynamical simulations (“too massive, too early”) has been read as possible evidence for new physics. We test this by confronting IllustrisTNG (TNG100-1) cumulative number densities $n(> M_\star, z)$ against JWST-era observations on a like-for-like stellar-mass basis. At $z \simeq 5\text{--}6$ the observed $n(> 10^{10.5} M_\odot)$ (Weibel et al. 2024, $\approx 3 \times 10^{-5} \text{ Mpc}^{-3}$) exceeds TNG ($1.1 \times 10^{-5} \text{ Mpc}^{-3}$ at $z = 5$) by a factor ≈ 2.7 ; but given the steep massive-end slope of the simulated stellar mass function ($d \log n / d \log M_\star \approx -1.6$), this excess is fully erased by a downward stellar-mass shift of only ≈ 0.28 dex — well inside the stellar-mass systematic budget, which the recent literature places near ~ 1 dex once SED-fitting priors, IMF assumptions, nebular-emission and AGN (“Little Red Dot”) contamination, and Eddington bias are included. We therefore find no robust tension at $z \simeq 5\text{--}6$. The larger apparent excess at $z \simeq 7\text{--}9$ (Labé et al. 2023 candidates, $\approx 13\times$) requires ≈ 0.44 dex — still within a plausible budget — but rests on unconfirmed photometric masses. We flag as a distinct, harder residual the spectroscopically confirmed quiescent galaxies at $z > 6$, whose ~ 2 dex excess is not dissolved by these systematics. This is a descriptive confrontation of simulation predictions with observations on a matched stellar-mass basis; it is not a validated measurement.

1. INTRODUCTION

Early JWST photometry reported an unexpected abundance of luminous, apparently very massive galaxies at $z > 6$, implying baryon-to-stellar conversion efficiencies approaching or exceeding unity when compared to the Λ CDM halo mass function (Labé et al. 2023). Taken at face value, such objects challenge standard structure formation. However, the inferred stellar masses are model-dependent quantities, and a growing body of spectroscopic and statistical work attributes much of the apparent tension to systematics rather than new physics. Here we ask a single, falsifiable question: *does the observed massive-galaxy abundance exceed the IllustrisTNG prediction by more than the stellar-mass systematic budget?*

2. DATA AND METHOD

We take cumulative comoving number densities $n(> M_\star, z)$ from the public IllustrisTNG (TNG100-1) subhalo catalogs, and compare to two observed anchors of $n(> 10^{10.5} M_\odot)$: Weibel et al. (2024) at $z \simeq 5\text{--}6$ and Labé et al. (2023) candidates at $z \simeq 7\text{--}9$. The comparison is non-circular by construction: TNG masses are predictions, the observations are independent. We quantify the tension as the downward stellar-mass shift Δ required to bring the observed $n(> 10^{10.5})$ into agree-

ment with TNG, using the simulated massive-end slope $d \log n / d \log M_\star$ measured between $\log M_\star = 10.0$ and 10.5 .

3. RESULTS

At $z = 5$, TNG gives $n(> 10^{10.5}) = 1.1 \times 10^{-5} \text{ Mpc}^{-3}$ and the observed value is 3×10^{-5} , a factor 2.7 (0.43 dex). With a local slope $d \log n / d \log M_\star \approx -1.58$, the excess is erased by $\Delta \approx 0.28$ dex. At $z \simeq 7\text{--}9$ the apparent excess is $\approx 13.6\times$ (1.13 dex) and requires $\Delta \approx 0.44$ dex. The stellar-mass systematic budget from independent recent analyses is substantially larger than either: SED-fitting codes disagree by ~ 1 dex on identical photometry with statistically indistinguishable fits; a top-heavy IMF alone lowers $M_\star$ by up to 1.0 dex; removing AGN continuum in broad-line “Little Red Dots” reduces host masses by orders of magnitude; and Eddington bias convolved with the steep halo mass function amplifies the apparent high-mass abundance, dropping the required efficiency to $\epsilon \simeq 0.2\text{--}0.4$.

Table 1 generalizes this single point: the downward stellar-mass shift required to erase the excess stays within the ~ 1 dex budget across the full plausible range of observed excess ($2\text{--}20\times$) and simulated massive-end slope (-1.4 to -2.0). Only a genuine ~ 2 dex excess

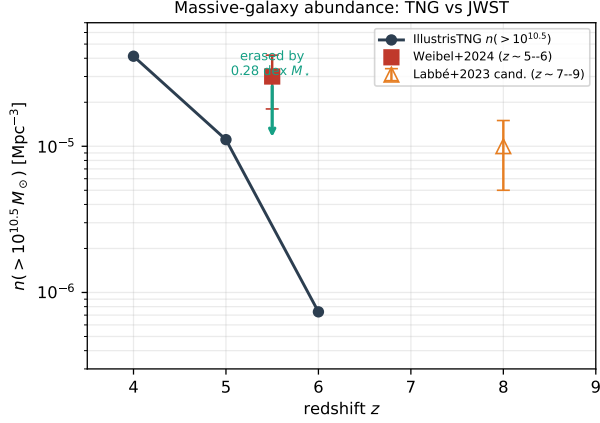


Figure 1. Cumulative number density $n(> 10^{10.5} M_{\odot})$ vs redshift. IllustrisTNG (dark line/circles) compared to observed anchors (Weibel+2024 at $z \simeq 5-6$; Labé+2023 candidates at $z \simeq 7-9$). The $z \simeq 5-6$ excess over TNG is erased by a 0.28 dex downward stellar-mass shift, well within the ~ 1 dex systematic budget (green arrow).

— the regime of the spectroscopic quiescent galaxies — would require $\gtrsim 1.4$ dex and exceed it.

Table 1. Downward stellar-mass shift $\Delta \log M_{\star}$ (dex) required to erase the observed massive-galaxy abundance excess over IllustrisTNG, across the observed excess factor and the massive-end simulated SMF slope $s = d \log n / d \log M_{\star}$. Entries ≤ 1 dex lie within the stellar-mass systematic budget.

Observed excess	$s=-1.4$	$s=-1.6$	$s=-1.8$	$s=-2.0$
$2\times$ (conservative)	0.22	0.19	0.17	0.15
$2.7\times$ ($z \simeq 5-6$)	0.31	0.27	0.24	0.22
$13\times$ ($z \simeq 7-9$)	0.80	0.70	0.62	0.56
$20\times$ (extreme)	0.93	0.81	0.72	0.65

4. DISCUSSION

The $z \simeq 5-6$ excess is comfortably within the systematic budget and we regard it as consistent with TNG. The $z \simeq 7-9$ excess, while larger, both sits within a plausible budget and rests on photometric candidate masses that spectroscopy has repeatedly revised downward. A genuine, distinct residual remains: the spectroscopically confirmed compact quiescent galaxies at $z > 6$, whose inferred number density lies ~ 2 dex above baseline predictions and is not dissolved by the star-forming systematics above; this warrants dedicated follow-up and is not claimed to be resolved here.

5. CONCLUSION

The “too massive, too early” tension between JWST observations and IllustrisTNG at $z \simeq 4-6$ is not robust to the stellar-mass systematic budget. We do not claim a measurement of consistency, only that the data do not require a departure from Λ CDM at these redshifts once realistic mass systematics are allowed. This is a descriptive, automated confrontation; it has not been validated by human review.

REFERENCES

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